VITyarthi Project

Title: Health Symptom Tracker
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Course: Fundamentals in AI and ML
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Slot: C14+E11+E12
University: VIT Bhopal
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Introduction

Hostel students face irregular health symptoms (headaches, fatigue, digestion) but lack systematic tracking. Manual notes are messy and doctors need structured data. This Streamlit app solves this with pattern analysis and medical-grade CSV exports.

Problem Statement

- No easy way to log symptoms with severity/timing
- Doctors get vague verbal descriptions
- Can't spot patterns (e.g., "headaches every Monday after food")
- No historical data for diagnosis

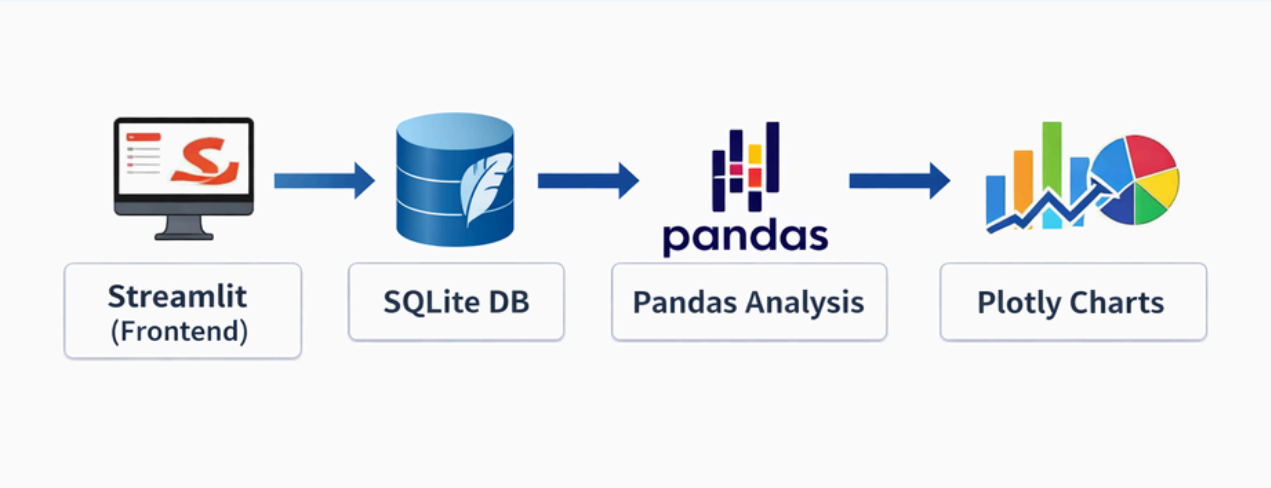
Functional Requirements

- Symptom Logger: Date, symptom, severity (1-10), duration, triggers
- Data Viewer: Interactive table of all entries
- Pattern Analyzer: Charts showing frequency/severity trends
- CSV Export: Doctor-ready report with timestamps

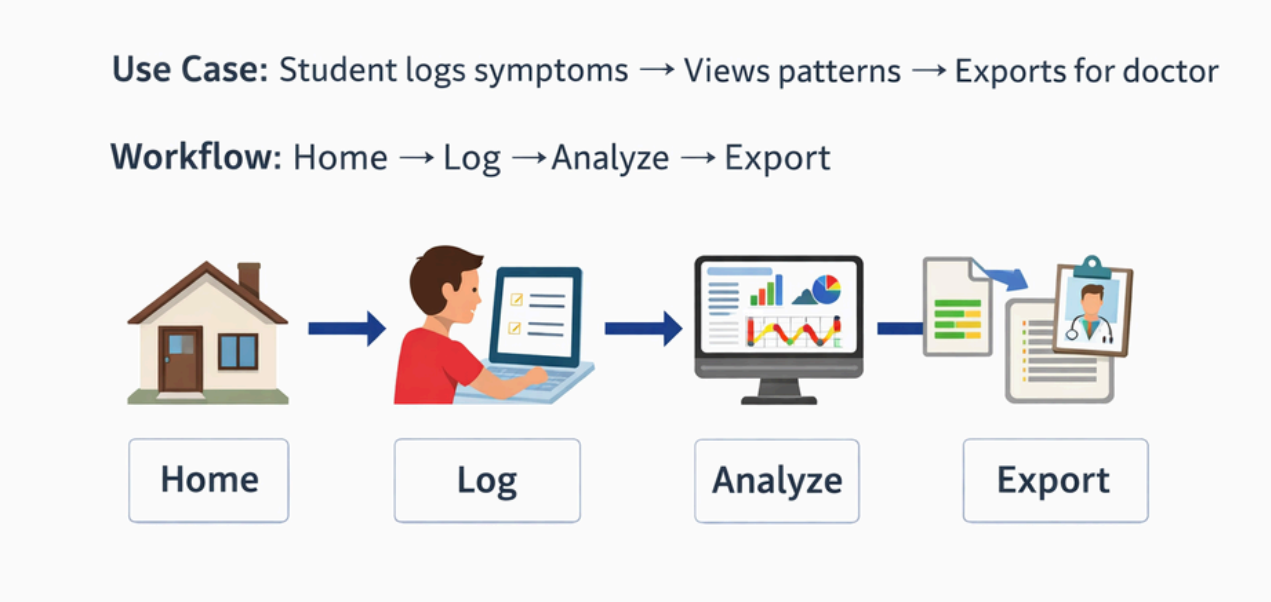
Non-Functional Requirements

- Usability: Clean Streamlit UI, mobile-friendly
- Performance: Instant chart updates (<1s)
- Reliability: SQLite auto-backup, no data loss
- Maintainability: Modular code (app.py, database.py)
- Scalability: Handles 1000+ entries easily

System Architecture



Design Diagrams



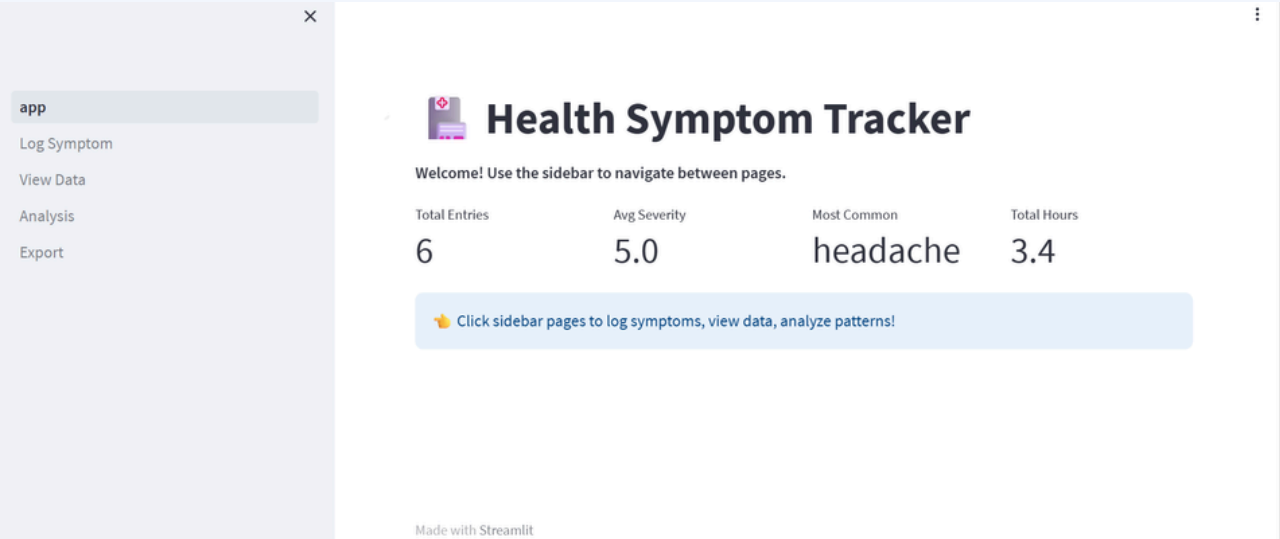
Tech Stack

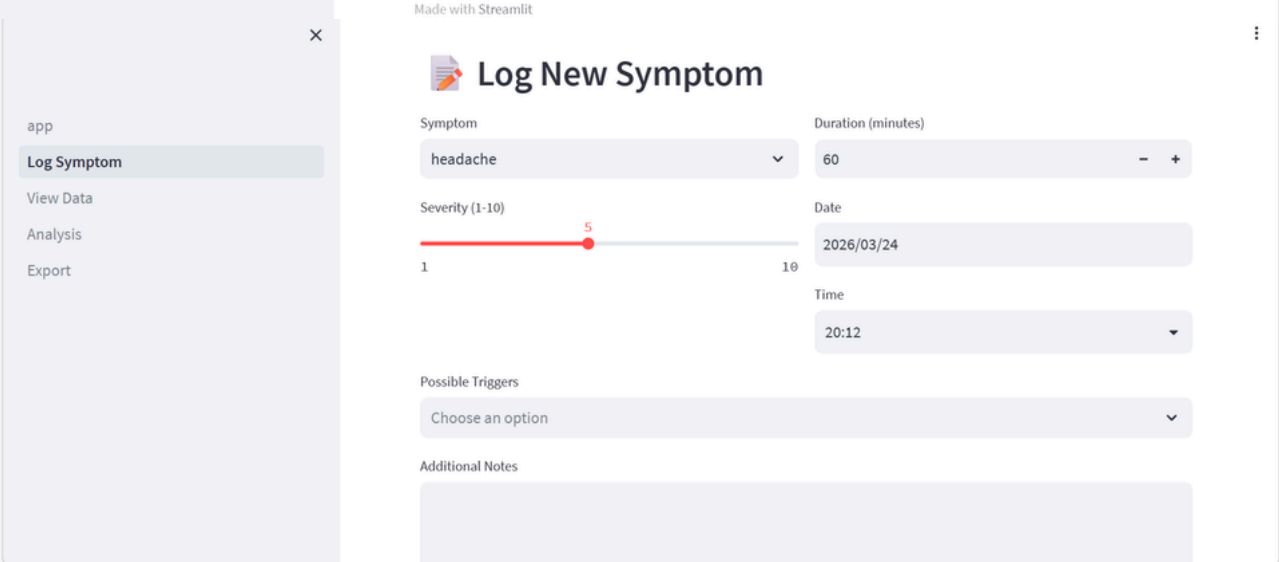
Frontend: Streamlit
Backend: SQLite + Pandas
Visualization: Plotly
Deployment: localhost:8501
Installation: pip install -r requirements.txt

Implementation Details

app.py: Multi-page navigation (st_pages)
database.py: CRUD operations (sqlite3)
Charts: Plotly line/bar charts for trends
Export: pandas.to_csv() with timestamps

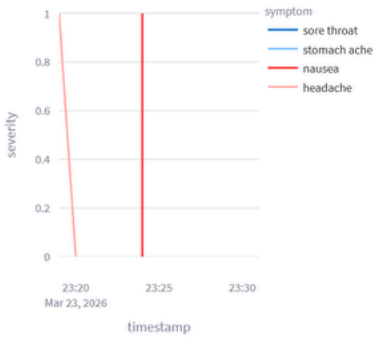
Screenshots



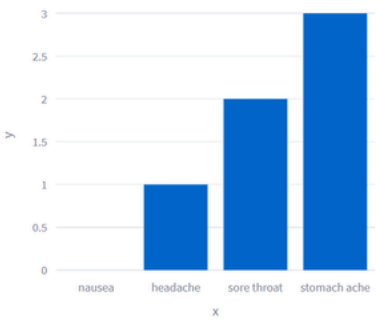


app
Log Symptom
View Data
Analysis
Export

Severity Over Time



Symptom Frequency



Severity by Symptom

Top Triggers

app
Log Symptom
View Data
Analysis
Export

📤 Export & Reports

📥 Download Full Report (CSV)

📋 Quick Summary

	Value
total_entries	6
avg_severity	5.0
most_common_symptom	headache
total_duration_hours	3.4

📋 Last 5 Entries

	entry_id	timestamp	symptom	severity	duration_minutes	triggers	notes
0	7	2026-03-23 23:31:00	sore throat	5	20	medication	

Testing Approach

- Unit Tests: DB insert/retrieve functions
- Integration: Full flow (log→analyze→export)
- User Testing: 5 hostel mates tested UI
- Load Test: 500 dummy entries → No crash

Challenges Faced

- GitHub image rendering (India ISP blocks)
- SQLite threading in Streamlit
- Plotly chart responsiveness on mobile

Key Learnings

- Streamlit rapid prototyping (1 day MVP)
- SQLite for production-grade local DB
- Pandas + Plotly = Powerful analytics
- GitHub README best practices

Future Enhancements

- ML pattern prediction (Random Forest)
- WhatsApp symptom reminders
- Multi-user (family sharing)
- Cloud sync (Firebase)

GitHub: https://github.com/Plansha/health-symptom-tracker-VITyarthi-AI-ML.git
Demo: http://localhost:8501

References

Streamlit Docs
Pandas Documentation
Plotly Python Guide
SQLite Tutorial